

ON A HIGH-ORDER ITERATIVE SCHEME FOR A NONLINEAR LOVE EQUATION

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Abstract. In this paper, a high-order iterative scheme is established for a nonlinear Love equation associated with homogeneous Dirichlet boundary conditions. This is a development based on recent results (L. T. P. Ngoc, N. T. Long (2011); L. X. Truong, L. T. P. Ngoc, N. T. Long (2009)) to get a convergent sequence at a rate of order $N \geq 2$ to a local unique weak solution of the above mentioned equation.

Keywords: nonlinear Love equation; Faedo-Galerkin method; convergence of order N

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1. INTRODUCTION

In this paper, we consider the following Dirichlet problem for a nonlinear Love equation

$$(1.1) \quad u_{tt} - u_{xx} - u_{xxtt} = f(x, t, u), \quad 0 < x < 1, \quad 0 < t < T,$$

$$(1.2) \quad u(0, t) = u(1, t) = 0,$$

$$(1.3) \quad u(x, 0) = \tilde{u}_0(x), \quad u_t(x, 0) = \tilde{u}_1(x),$$

where $\tilde{u}_0, \tilde{u}_1, f$ are given functions.

When $f = 0$, (1.1) is related to the Love equation

$$(1.4) \quad u_{tt} - \frac{E}{\varrho} u_{xx} - 2\mu^2 k^2 u_{xxtt} = 0,$$

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presented by V. Radochová in 1978 (see [17]). This equation describes the vertical oscillations of a rod, which was established from Euler's variational equation of an energy functional

$$(1.5) \quad \int_0^T dt \int_0^L \left[\frac{1}{2} F \varrho (u_t^2 + \mu^2 k^2 u_{tx}^2) - \frac{1}{2} F (E u_x^2 + \varrho \mu^2 k^2 u_x u_{xtt}) \right] dx.$$

The parameters in (1.5) have the following meaning: u is the displacement, L is the length of the rod, F is the area of cross-section, k is the cross-section radius, E is the Young modulus of the material, and ϱ is the mass density. By using the Fourier method, Radochová [17] obtained a classical solution of equation (1.4) associated with the initial condition (1.3) and boundary conditions

$$u(0, t) = u(L, t) = 0,$$

or

$$(1.6) \quad \begin{cases} u(0, t) = 0, \\ \varepsilon u_{xtt}(L, t) + c^2 u_x(L, t) = 0, \end{cases}$$

where $c^2 = E/\varrho$, $\varepsilon = 2\mu^2 k^2$.

Equations of Love waves or Love-type waves have been studied by many authors, we refer to [3], [6], [11], [10], [16], and references therein.

In [10], by combining the linearization method for the nonlinear term, the Faedo-Galerkin method and the weak compactness method, the existence of a unique weak solution of a Dirichlet problem for the nonlinear Love equation $u_{tt} - u_{xx} - u_{xxtt} = f(x, t, u, u_x, u_t, u_{xt})$ is proved. We note, however, the recurrent sequence obtained here converges only at a rate of order 1.

It is well known that Newton's method and its variants are used to solve nonlinear operator equations or systems of nonlinear equations, see [15] and references therein. In case $\lim_{n \rightarrow \infty} u_n = u$, one speaks of *convergence of order N* if $|u_{n+1} - u| \leq C|u_n - u|^N$ for some $C > 0$ and all large N . In the special cases $N = 1$ with $C < 1$ and $N = 2$ one also speaks of linear and quadratic convergence, respectively, see [5]. Based on the ideas about recurrence relations of these methods, a high-order iterative scheme can be constructed for solving the nonlinear operator equation, see [13], [12], [19], [20].

In [18], a symmetric version of the regularized long wave equation (SRLWE)

$$(1.7) \quad \begin{cases} u_{xxt} - u_t = \varrho_x + uu_x, \\ \varrho_t + u_x = 0, \end{cases}$$

has been proposed as a model for propagation of weakly nonlinear ion acoustic and space-charge waves. Obviously, eliminating ϱ from (1.7), we get

$$(1.8) \quad u_{tt} - u_{xx} - u_{xxtt} = -uu_{xt} - u_x u_t.$$

The SRLWE (1.8) is explicitly symmetric in the x and t derivatives and it is very similar to the regularized long wave equation which describes shallow water waves and plasma drift waves [1], [2]. The SRLWE also arises in many other areas of mathematical physics [4], [9], [14]. Note that (1.8) is a special form of the equation discussed in [10].

Motivated by results for Love equations in [11], [10], and based on the use of a high-order iterative scheme in [13], [12], [19], [20], in this note, we will establish a similar scheme to get the convergence of order N for problem (1.1)–(1.3). To achieve this purpose, we define a recurrent sequence $\{u_m\}$ associated with equation (1.1) as follows:

$$\frac{\partial^2 u_m}{\partial t^2} - \frac{\partial^2 u_m}{\partial x^2} - \frac{\partial^4 u_m}{\partial t^2 \partial x^2} = \sum_{i=0}^{N-1} \frac{1}{i!} \frac{\partial^i f}{\partial u^i}(x, t, u_{m-1})(u_m - u_{m-1})^i, \quad 0 < x < 1, \quad 0 < t < T,$$

with u_m satisfying (1.2), (1.3) and $u_0 \equiv 0$. If $f \in C^N([0, 1] \times \mathbb{R}_+ \times \mathbb{R})$, we prove that the sequence $\{u_m\}$ converges at a rate of order N to a weak unique solution of problem (1.1)–(1.3).

Note that, if equation (1.1) does not contain the term u_{xxtt} , a solution u of problem (1.1)–(1.3) can be found in the space $S_1 = \{u \in L^\infty(0, T; H_0^1 \cap H^2) : u_t \in L^\infty(0, T; H_0^1), u_{tt} \in L^\infty(0, T; L^2)\}$, whereas adding the term u_{xxtt} yields $u \in S = \{u \in L^\infty(0, T; H_0^1 \cap H^2) : u_t, u_{tt} \in L^\infty(0, T; H_0^1 \cap H^2)\}$. Since $S \subset S_1$, it means that the regularity of solutions improves.

2. A HIGH-ORDER ITERATIVE SCHEME

First, we put $\Omega = (0, 1)$ and denote the usual function spaces used in this paper by $L^p = L^p(\Omega)$, $H^m = H^m(\Omega)$. Let $\langle \cdot, \cdot \rangle$ be either the scalar product in L^2 or the dual pairing of a continuous linear functional and an element of a function space. The notation $\|\cdot\|$ stands for the norm in L^2 and we denote by $\|\cdot\|_X$ the norm in the Banach space X . We call X' the dual space of X .

Let $u(t)$, $u'(t) = u_t(t) = \dot{u}(t)$, $u''(t) = u_{tt}(t) = \ddot{u}(t)$, $u_x(t) = \nabla u(t)$, $u_{xx}(t) = \Delta u(t)$, denote $u(x, t)$, $(\partial u / \partial t)(x, t)$, $(\partial^2 u / \partial t^2)(x, t)$, $(\partial u / \partial x)(x, t)$, $(\partial^2 u / \partial x^2)(x, t)$, respectively.

Next, we will define the following norms on appropriate spaces. This functional setting allows us to make precise the concept of a weak solution of problem (1.1)–(1.3) used in this note. We will use the norm $\|v\|_{H^1} = (\|v\|^2 + \|v_x\|^2)^{1/2}$ on H^1 . It is known that the imbedding $H^1 \hookrightarrow C^0([0, 1])$ is compact and $\|v\|_{C^0([0, 1])} \leq \sqrt{2}\|v\|_{H^1}$, for all $v \in H^1$. Furthermore, on $H_0^1 = \{u \in H^1: u(0) = u(1) = 0\}$, the two norms $v \mapsto \|v\|_{H^1}$ and $v \mapsto \|v_x\|$ are equivalent and $\|v\|_{C^0([0, 1])} \leq \|v_x\|$ for all $v \in H_0^1$. Finally, on $H_0^1 \cap H^2 = \{v \in H^2: v(0) = v(1) = 0\}$, we will use the norm $\|v\|_{H_0^1 \cap H^2} = \sqrt{\|v_x\|^2 + \|v_{xx}\|^2}$.

Definition. We say that u is a weak solution of problem (1.1)–(1.3) if

$$u \in L^\infty(0, T; H_0^1 \cap H^2), \quad \dot{u}, \ddot{u} \in L^\infty(0, T; H_0^1 \cap H^2),$$

and u satisfies the following variational equation:

$$\langle \ddot{u}(t), w \rangle + \langle u_x(t) + \ddot{u}_x(t), w_x \rangle = \langle f(x, t, u), w \rangle$$

for all $w \in H_0^1$ and a.e. $t \in (0, T)$, together with the initial conditions

$$u(0) = \tilde{u}_0, \quad \dot{u}(0) = \tilde{u}_1.$$

Now, we make the following assumptions:

(A₁) $\tilde{u}_0, \tilde{u}_1 \in H_0^1 \cap H^2$,

(A₂) $f \in C^1([0, 1] \times \mathbb{R}_+ \times \mathbb{R})$ such that

(i) $\partial^i f / \partial u^i \in C^1([0, 1] \times \mathbb{R}_+ \times \mathbb{R})$, $0 \leq i \leq N - 1$,

(ii) $\partial^N f / \partial u^N \in C^0([0, 1] \times \mathbb{R}_+ \times \mathbb{R})$,

(iii) $f(0, t, 0) = f(1, t, 0) = 0$ for all $t \geq 0$.

Fix $T^* > 0$. For each $M > 0$ given, we set the constants $K_0(M, f)$, $K_1(M, f)$, $K_M(f)$ as follows:

$$\begin{cases} K_0(M, f) = \sup\{|f(x, t, u)|: 0 \leq x \leq 1, 0 \leq t \leq T^*, |u| \leq M\}, \\ K_1(M, f) = K_0(M, f) + K_0\left(M, \frac{\partial f}{\partial x}\right) + K_0\left(M, \frac{\partial f}{\partial t}\right) + K_0\left(M, \frac{\partial f}{\partial u}\right), \\ K_M(f) = \sum_{i=0}^{N-1} K_1\left(M, \frac{\partial^i f}{\partial u^i}\right) + K_0\left(M, \frac{\partial^N f}{\partial u^N}\right). \end{cases}$$

For every $T \in (0, T^*]$ and $M > 0$, we put

$$\begin{cases} W(M, T) = \{v \in L^\infty(0, T; H_0^1 \cap H^2): v_t \in L^\infty(0, T; H_0^1 \cap H^2), \\ \quad v_{tt} \in L^\infty(0, T; H_0^1), \text{ with } \|v\|_{L^\infty(0, T; H_0^1 \cap H^2)}, \\ \quad \|v_t\|_{L^\infty(0, T; H_0^1 \cap H^2)}, \|v_{tt}\|_{L^\infty(0, T; H_0^1)} \leq M\}, \\ W_1(M, T) = \{v \in W(M, T): v_{tt} \in L^\infty(0, T; H_0^1 \cap H^2)\}. \end{cases}$$

In the following, we will establish the recurrent sequence $\{u_m\}$ via a high-order iterative scheme.

Theorem 2.1. *Suppose that the assumptions (A_1) , (A_2) are fulfilled. Then there exist positive constants M, T and a sequence $\{u_m\} \subset W_1(M, T)$ defined as follows:*

- (i) *the first term is $u_0 = 0$;*
- (ii) *with each given term*

$$(2.1) \quad u_{m-1} \in W_1(M, T),$$

there exists $u_m \in W_1(M, T)$ ($m \geq 1$) satisfying

$$(2.2) \quad \begin{cases} \langle \ddot{u}_m(t), w \rangle + \langle u_{mx}(t) + \dot{u}_{mx}(t), w_x \rangle = \langle F_m(t), w \rangle & \forall w \in H_0^1, \\ u_m(0) = \tilde{u}_0, \dot{u}_m(0) = \tilde{u}_1, \end{cases}$$

in which

$$(2.3) \quad F_m(x, t) = \sum_{i=0}^{N-1} \frac{1}{i!} \frac{\partial^i f}{\partial u^i}(x, t, u_{m-1})(u_m - u_{m-1})^i.$$

Proof. Approximating solutions. To prove this theorem, we use the Faedo-Galerkin method.

Consider a special orthonormal basis $\{w_j\}$ on H_0^1 : $w_j(x) = \sqrt{2} \sin(j\pi x)$, $j = 1, 2, \dots$, formed by the eigenfunctions of the Laplacian $-\Delta = -\partial^2/\partial x^2$. It is clear that w_j satisfies

$$-\Delta w_j = \lambda_j w_j, \quad w_j \in H_0^1 \cap H^2, \quad \lambda_j = (j\pi)^2, \quad j = 1, 2, \dots$$

If

$$u_m^{(k)}(t) = \sum_{j=1}^k c_{mj}^{(k)}(t) w_j$$

is a solution of the system

$$(2.4) \quad \begin{cases} \langle \ddot{u}_m^{(k)}(t), w_j \rangle + \langle u_{mx}^{(k)}(t) + \dot{u}_{mx}^{(k)}(t), w_{jx} \rangle = \langle F_m^{(k)}(t), w_j \rangle, & j = 1, 2, \dots, k, \\ u_m^{(k)}(0) = \tilde{u}_{0k}, \dot{u}_m^{(k)}(0) = \tilde{u}_{1k}, \end{cases}$$

with

$$(2.5) \quad \begin{cases} \tilde{u}_{0k} = \sum_{j=1}^k \alpha_j^{(k)} w_j \longrightarrow \tilde{u}_0 & \text{strongly in } H_0^1 \cap H^2, \\ \tilde{u}_{1k} = \sum_{j=1}^k \beta_j^{(k)} w_j \longrightarrow \tilde{u}_1 & \text{strongly in } H_0^1 \cap H^2, \end{cases}$$

and

$$(2.6) \quad \left\{ \begin{array}{l} F_m^{(k)}(x, t) = \sum_{i=0}^{N-1} \frac{1}{i!} \frac{\partial^i f}{\partial u^i}(x, t, u_{m-1})(u_m^{(k)} - u_{m-1})^i \\ \quad = \sum_{j=0}^{N-1} \Psi_j(x, t, u_{m-1})(u_m^{(k)})^j, \\ \Psi_j(x, t, u_{m-1}) = \sum_{i=j}^{N-1} \frac{(-1)^{i-j}}{j!(i-j)!} \frac{\partial^i f}{\partial u^i}(x, t, u_{m-1}) u_{m-1}^{i-j}, \end{array} \right.$$

then $c_{mj}^{(k)}$ satisfies the following system of nonlinear ordinary differential equations:

$$(2.7) \quad \left\{ \begin{array}{l} \ddot{c}_{mj}^{(k)}(t) + \mu_j^2 c_{mj}^{(k)}(t) = f_{mj}^{(k)}(t), \\ c_{mj}^{(k)}(0) = \alpha_j^{(k)}, \quad \dot{c}_{mj}^{(k)}(0) = \beta_j^{(k)}, \quad 1 \leq j \leq k, \end{array} \right.$$

where

$$f_{mj}^{(k)}(t) = \frac{1}{1 + \lambda_j} \langle F_m^{(k)}(t), w_j \rangle, \quad \mu_j^2 = \frac{\lambda_j}{1 + \lambda_j}, \quad \lambda_j = (j\pi)^2, \quad 1 \leq j \leq k.$$

Using Banach's contraction principle, it is not difficult to show that (2.7) has a unique solution $c_{mj}^{(k)}(t)$ in $[0, T_m^{(k)}]$, with certain $T_m^{(k)} \in (0, T]$ (see [12]). Therefore, (2.4) has a unique solution $u_m^{(k)}(t)$ in $[0, T_m^{(k)}]$.

The following estimates allow one to take $T_m^{(k)} = T$ independent of m and k . By such a priori estimates of $u_m^{(k)}(t)$, it can be extended outside $[0, T_m^{(k)}]$ and then, a solution defined in $[0, T]$ will be obtained.

Estimates. Multiply (2.4)₁ by $\dot{c}_{mj}^{(k)}(t)$ and sum over j . After that, integrating with respect to the time variable from 0 to t , we have

$$(2.8) \quad \begin{aligned} p_m^{(k)}(t) &\equiv \|\dot{u}_m^{(k)}(t)\|^2 + \|u_{mx}^{(k)}(t)\|^2 + \|\dot{u}_{mx}^{(k)}(t)\|^2 \\ &= p_m^{(k)}(0) + 2 \int_0^t \langle F_m^{(k)}(s), \dot{u}_m^{(k)}(s) \rangle ds. \end{aligned}$$

Replacing w_j in (2.4)₁ by $-w_{jxx}/\lambda_j$, and integrating by parts, we obtain

$$\langle \ddot{u}_{mx}^{(k)}(t), w_{jx} \rangle + \langle u_{mxx}^{(k)}(t) + \ddot{u}_{mxx}^{(k)}(t), w_{jxx} \rangle = \langle F_{mx}^{(k)}(t), w_{jx} \rangle, \quad 1 \leq j \leq k,$$

therefore, in the same way as (2.8),

$$(2.9) \quad \begin{aligned} q_m^{(k)}(t) &\equiv \|\dot{u}_{mx}^{(k)}(t)\|^2 + \|u_{mxx}^{(k)}(t)\|^2 + \|\dot{u}_{mxx}^{(k)}(t)\|^2 \\ &= q_m^{(k)}(0) + 2 \int_0^t \langle F_{mx}^{(k)}(s), \dot{u}_{mx}^{(k)}(s) \rangle ds. \end{aligned}$$

Furthermore, because $c_{mj}^{(k)}(t)$ is a solution of the system (2.7), both $\ddot{c}_{mj}^{(k)}(t)$ and $\ddot{u}_m^{(k)}(t) = \sum_{j=1}^k \ddot{c}_{mj}^{(k)}(t)w_j$ are defined. Hence, we can take the derivative with respect to t of (2.4)₁ and then

$$(2.10) \quad \langle \ddot{u}_m^{(k)}(t), w_j \rangle + \langle \dot{u}_{mx}^{(k)}(t) + \ddot{u}_{mx}^{(k)}(t), w_{jx} \rangle = \langle \dot{F}_m^{(k)}(t), w_j \rangle,$$

for all $1 \leq j \leq m$. Multiplying (2.10) by $\ddot{c}_{mj}(t)$, summing over j and integrating from 0 to t implies

$$(2.11) \quad \begin{aligned} r_m^{(k)}(t) &= \|\ddot{u}_m^{(k)}(t)\|^2 + \|\dot{u}_{mx}^{(k)}(t)\|^2 + \|\ddot{u}_{mx}^{(k)}(t)\|^2 \\ &= r_m^{(k)}(0) + 2 \int_0^t \langle \dot{F}_m^{(k)}(s), \ddot{u}_m^{(k)}(s) \rangle ds. \end{aligned}$$

Combining (2.8), (2.9), and (2.11) leads to

$$(2.12) \quad \begin{aligned} S_m^{(k)}(t) &= p_m^{(k)}(t) + q_m^{(k)}(t) + r_m^{(k)}(t) \\ &= S_m^{(k)}(0) + 2 \int_0^t \langle F_m^{(k)}(s), \dot{u}_m^{(k)}(s) \rangle ds \\ &\quad + 2 \int_0^t \langle F_{mx}^{(k)}(s), \dot{u}_{mx}^{(k)}(s) \rangle ds + 2 \int_0^t \langle \dot{F}_m^{(k)}(s), \ddot{u}_m^{(k)}(s) \rangle ds \\ &\equiv S_m^{(k)}(0) + \sum_{j=1}^3 I_j. \end{aligned}$$

Letting $t \rightarrow 0_+$ in (2.4)₁ and multiplying the result obtained by $\ddot{c}_{mj}^{(k)}(0)$, we get

$$\|\ddot{u}_m^{(k)}(0)\|^2 + \|\ddot{u}_{mx}^{(k)}(0)\|^2 + \langle u_{mx}^{(k)}(0), \ddot{u}_{mx}^{(k)}(0) \rangle = \langle F_m^{(k)}(0), \ddot{u}_m^{(k)}(0) \rangle.$$

Consequently,

$$\begin{aligned} \xi_m^{(k)} &= \|\ddot{u}_m^{(k)}(0)\|^2 + \|\ddot{u}_{mx}^{(k)}(0)\|^2 \\ &\leq \|u_{mx}^{(k)}(0)\| \|\ddot{u}_{mx}^{(k)}(0)\| + \|F_m^{(k)}(0)\| \|\ddot{u}_m^{(k)}(0)\| \\ &\leq \|u_{mx}^{(k)}(0)\| \sqrt{\xi_m^{(k)}} + \|F_m^{(k)}(0)\| \sqrt{\xi_m^{(k)}} \\ &\leq \frac{1}{2} \xi_m^{(k)} + \frac{1}{2} (\|u_{mx}^{(k)}(0)\| + \|F_m^{(k)}(0)\|)^2 \\ &= \frac{1}{2} \xi_m^{(k)} + \frac{1}{2} \left(\|\tilde{u}_{0kx}\| + \left\| \sum_{i=0}^{N-1} \frac{1}{i!} \frac{\partial^i f}{\partial u^i}(x, t, \tilde{u}_0) (\tilde{u}_{0k} - \tilde{u}_0)^i \right\| \right)^2 \\ &= \frac{1}{2} \xi_m^{(k)} + \frac{1}{2} \left(\|\tilde{u}_{0kx}\| + \sum_{i=0}^{N-1} \frac{(\|\tilde{u}_{0kx}\| + \|\tilde{u}_{0x}\|)^i}{i!} \sup_{\substack{0 \leq x \leq 1, 0 \leq t \leq T^*, \\ |z| \leq \|\tilde{u}_{0x}\|}} \left| \frac{\partial^i f}{\partial u^i}(x, t, z) \right| \right)^2, \end{aligned}$$

which gives that for all $m, k \in \mathbb{N}$,

$$(2.13) \quad \xi_m^{(k)} \leq \left(\|\tilde{u}_{0kx}\| + \sum_{i=0}^{N-1} \frac{(\|\tilde{u}_{0kx}\| + \|\tilde{u}_{0x}\|)^i}{i!} \sup_{\substack{0 \leq x \leq 1, 0 \leq t \leq T^*, \\ |z| \leq \|\tilde{u}_{0x}\|}} \left| \frac{\partial^i f}{\partial u^i}(x, t, z) \right| \right)^2.$$

By (2.5) and (2.13), we can deduce that there exists a constant $M > 0$, independent of k and m , such that

$$(2.14) \quad \begin{aligned} S_m^{(k)}(0) &= \|\tilde{u}_{1k}\|^2 + \|\tilde{u}_{0k}\|^2 + 3\|\tilde{u}_{1kx}\|^2 + \|\tilde{u}_{0kxx}\|^2 + \|\tilde{u}_{1kxx}\|^2 + \xi_m^{(k)} \\ &\leq \frac{M^2}{4} \quad \forall m, k \in \mathbb{N}. \end{aligned}$$

In order to continue the proof, we will state the following properties of $F_m^{(k)}(t)$, $F_{mx}^{(k)}(t)$, $\dot{F}_m^{(k)}(t)$. Their proof is analogous to [12], Lemma 3.3.

$$(2.15) \quad \begin{aligned} \text{(i)} \quad & \|F_m^{(k)}(t)\| \leq \tilde{b}_M \left[1 + \left(\sqrt{S_m^{(k)}(t)} \right)^{N-1} \right], \\ \text{(ii)} \quad & \|F_{mx}^{(k)}(t)\| \leq \tilde{b}_M \left[1 + \left(\sqrt{S_m^{(k)}(t)} \right)^{N-1} \right], \\ \text{(iii)} \quad & \|\dot{F}_m^{(k)}(t)\| \leq \tilde{b}_M \left[1 + \left(\sqrt{S_m^{(k)}(t)} \right)^{N-1} \right], \end{aligned}$$

where $\tilde{b}_M = (M + N)K_M(f) \sum_{i=0}^{N-1} \tilde{a}_i$ and $\tilde{a}_0 = 1 + \sum_{i=1}^{N-1} 2^{i-1} M^i / i!$, $\tilde{a}_i = 2^{i-1} / i!$, $i = 1, 2, \dots, N-1$.

Using (2.15)(i), we have

$$(2.16) \quad \begin{aligned} I_1 &= 2 \int_0^t \langle F_m^{(k)}(s), \dot{u}_m^{(k)}(s) \rangle ds \leq 2 \int_0^t \|F_m^{(k)}(s)\| \|\dot{u}_m^{(k)}(s)\| ds \\ &\leq 2\tilde{b}_M \int_0^t \left[1 + \left(\sqrt{S_m^{(k)}(s)} \right)^{N-1} \right] \sqrt{S_m^{(k)}(s)} ds \leq 4\tilde{b}_M \left[T + \int_0^t (S_m^{(k)}(s))^N ds \right], \end{aligned}$$

and, similarly,

$$(2.17) \quad I_2 \leq 4\tilde{b}_M \left[T + \int_0^t (S_m^{(k)}(s))^N ds \right],$$

$$(2.18) \quad I_3 \leq 4\tilde{b}_M \left[T + \int_0^t (S_m^{(k)}(s))^N ds \right].$$

Combining (2.12), (2.14), (2.16)–(2.18), it follows that

$$(2.19) \quad S_m^{(k)}(t) \leq \frac{M^2}{4} + 12T\tilde{b}_M + 12\tilde{b}_M \int_0^t (S_m^{(k)}(s))^N ds, \quad 0 \leq t \leq T.$$

Then, by solving a nonlinear Volterra integral equation (based on the methods in [7]), it follows that there exists a constant $T > 0$ independent of k and m such that

$$(2.20) \quad S_m^{(k)}(t) \leq M^2 \quad \forall t \in [0, T], \quad \forall k, m \in \mathbb{N}.$$

Therefore, we can take constant $T_m^{(k)} = T$ for all m and k . Thus,

$$(2.21) \quad u_m^{(k)} \in W(M, T) \quad \text{for all } m \text{ and } k.$$

Convergence. Thanks to (2.21), there exists a subsequence of $\{u_m^{(k)}\}$, denoted by the same symbol, such that

$$(2.22) \quad \begin{cases} u_m^{(k)} \rightarrow u_m & \text{in } L^\infty(0, T; H_0^1 \cap H^2) \text{ weakly}^*, \\ \dot{u}_m^{(k)} \rightarrow \dot{u}_m & \text{in } L^\infty(0, T; H_0^1 \cap H^2) \text{ weakly}^*, \\ \ddot{u}_m^{(k)} \rightarrow \ddot{u}_m & \text{in } L^\infty(0, T; H_0^1) \text{ weakly}^*, \\ u_m \in W(M, T). \end{cases}$$

Applying the compactness lemma of Lions ([8], page 57) and the Riesz-Fischer theorem, from (2.22), there exists a subsequence of $\{u_m^{(k)}\}$, also denoted by the same symbol, satisfying

$$(2.23) \quad \begin{cases} u_m^{(k)} \rightarrow u_m & \text{strongly in } L^2(0, T; H_0^1) \text{ and a.e. in } Q_T, \\ \dot{u}_m^{(k)} \rightarrow \dot{u}_m & \text{strongly in } L^2(0, T; H_0^1) \text{ and a.e. in } Q_T. \end{cases}$$

On the other hand, by $L^\infty(0, T; H_0^1 \cap H^2) \hookrightarrow L^\infty(Q_T)$ and the inequality

$$|a^j - b^j| \leq jM^{j-1}|a - b| \quad \forall a, b \in [-M, M], \quad \forall M > 0, \quad \forall j \in \mathbb{N},$$

we deduce from (2.20) that

$$(2.24) \quad |(u_m^{(k)})^j - (u_m)^j| \leq jM^{j-1}|u_m^{(k)} - u_m|, \quad j = 0, \dots, N-1.$$

Therefore, (2.23) and (2.24) give

$$(2.25) \quad (u_m^{(k)})^j \rightarrow (u_m)^j \quad \text{strongly in } L^2(Q_T).$$

Note that

$$(2.26) \quad \begin{aligned} \|F_m^{(k)} - F_m\|_{L^2(Q_T)} &\leq \sum_{j=0}^{N-1} \|\Psi_j(\cdot, \cdot, u_{m-1})\|_{L^\infty(Q_T)} \|(u_m^{(k)})^j - (u_m)^j\|_{L^2(Q_T)} \\ &\leq K_M(f) \sum_{j=0}^{N-1} \sum_{i=j}^{N-1} \frac{M^{i-j}}{j!(i-j)!} \|(u_m^{(k)})^j - (u_m)^j\|_{L^2(Q_T)}, \end{aligned}$$

so (2.25) leads to

$$(2.27) \quad F_m^{(k)} \rightarrow F_m \quad \text{strongly in } L^2(Q_T).$$

Passing to limit in (2.4), (2.5), we have u_m satisfying (2.2), (2.3) in $L^2(0, T)$.
On the other hand, it follows from (2.2)₁ and (2.22)₄ that

$$\frac{\partial^2}{\partial x^2}(\ddot{u}_m(t) + u_m(t)) = \ddot{u}_m(t) - F_m(t) \in L^\infty(0, T; H_0^1).$$

Consequently,

$$\ddot{u}_m(t) + u_m(t) = \Phi \in L^\infty(0, T; H_0^1 \cap H^2),$$

and then

$$\ddot{u}_m(t) = \Phi - u_m(t) \in L^\infty(0, T; H_0^1 \cap H^2).$$

Hence, $u_m \in W_1(M, T)$ and Theorem 2.1 is proved. $\square$

Next, we set

$$W_1(T) = \{v \in L^\infty(0, T; H_0^1) : \dot{v} \in L^\infty(0, T; H_0^1)\}.$$

Then $W_1(T)$ is a Banach space with respect to the norm

$$\|v\|_{W_1(T)} = \|v\|_{L^\infty(0, T; H_0^1)} + \|\dot{v}\|_{L^\infty(0, T; H_0^1)}.$$

Theorem 2.2. *Suppose that the assumptions (A₁), (A₂) are fulfilled. Then*

- (i) *problem (1.1)–(1.3) has a unique weak solution $u \in W_1(M, T)$, where the constants $M > 0$ and $T > 0$ are chosen as in (2.14), (2.20).*

Furthermore,

- (ii) *the recurrent sequence $\{u_m\}$, defined by (2.1)–(2.3), converges at a rate of order N to the solution u strongly in the space $W_1(T)$ in the sense*

$$(2.28) \quad \|u_m - u\|_{W_1(T)} \leq C \|u_{m-1} - u\|_{W_1(T)}^N,$$

for all $m \geq 1$, where C is a suitable constant. On the other hand, the estimate is fulfilled

$$(2.29) \quad \|u_m - u\|_{W_1(T)} \leq C_T (\beta_T)^{N^m} \quad \forall m \in \mathbb{N},$$

where C_T and $\beta_T < 1$ are constants depending only on $\tilde{u}_0$, $\tilde{u}_1$, f , and T .

P r o o f. In the sequel, we will prove Theorem 2.2 only with $N \geq 2$.

Existence. We can prove that $\{u_m\}$ is a Cauchy sequence in $W_1(T)$.

Indeed, let $w_m = u_{m+1} - u_m$. Then w_m satisfies the variational problem

$$(2.30) \quad \begin{cases} \langle \ddot{w}_m(t), w \rangle + \langle w_{mx}(t) + \ddot{w}_{mx}(t), w_x \rangle = \langle F_{m+1}(t) - F_m(t), w \rangle & \forall w \in H_0^1, \\ w_m(0) = \dot{w}_m(0) = 0. \end{cases}$$

Taking $w = \dot{w}_m$ in (2.30), after integrating in t , we get

$$(2.31) \quad Z_m(t) = 2 \int_0^t \langle F_{m+1}(s) - F_m(s), \dot{w}_m(s) \rangle ds,$$

where

$$(2.32) \quad Z_m(t) = \|\dot{w}_m(t)\|^2 + \|w_{mx}(t)\|^2 + \|\dot{w}_{mx}(t)\|^2.$$

Using Taylor's expansion of the function $f(x, t, u_m)$ around the point u_{m-1} up to order N , we obtain

$$(2.33) \quad \begin{aligned} f(x, t, u_m) - f(x, t, u_{m-1}) \\ = \sum_{i=1}^{N-1} \frac{1}{i!} \frac{\partial^i f}{\partial u^i}(x, t, u_{m-1}) w_{m-1}^i + \frac{1}{N!} \frac{\partial^N f}{\partial u^N}(x, t, \bar{\lambda}_m) w_{m-1}^N, \end{aligned}$$

where $\bar{\lambda}_m = \bar{\lambda}_m(x, t) = u_{m-1} + \theta_1(u_m - u_{m-1})$, $0 < \theta_1 < 1$.

Hence, it follows from (2.3) and (2.33) that

$$\begin{aligned} F_{m+1}(x, t) - F_m(x, t) &= f(x, t, u_m) - f(x, t, u_{m-1}) \\ &+ \sum_{i=1}^{N-1} \frac{1}{i!} \frac{\partial^i f}{\partial u^i}(x, t, u_m) w_m^i - \sum_{i=1}^{N-1} \frac{1}{i!} \frac{\partial^i f}{\partial u^i}(x, t, u_{m-1}) w_{m-1}^i \\ &= \sum_{i=1}^{N-1} \frac{1}{i!} \frac{\partial^i f}{\partial u^i}(x, t, u_m) w_m^i + \frac{1}{N!} \frac{\partial^N f}{\partial u^N}(x, t, \bar{\lambda}_m) w_{m-1}^N. \end{aligned}$$

Thus, we have

$$(2.34) \quad \begin{aligned} \|F_{m+1}(t) - F_m(t)\| &\leq K_M(f) \sum_{i=1}^N \frac{1}{i!} \|w_{mx}(t)\|^i + \frac{1}{N!} K_M(f) \|w_{m-1}(t)\|^N \\ &\leq \gamma_T^{(1)} \sqrt{Z_m(t)} + \gamma_T^{(2)} (\sqrt{Z_{m-1}(t)})^N, \end{aligned}$$

where

$$\gamma_T^{(1)} = K_M(f) \sum_{i=1}^N \frac{1}{i!} M^{i-1}, \quad \gamma_T^{(2)} = \frac{1}{N!} K_M(f).$$

Then we deduce from (2.31), (2.32), and (2.34) that

$$(2.35) \quad \begin{aligned} Z_m(t) &\leq 2 \int_0^t \|F_{m+1}(s) - F_m(s)\| \|\dot{w}_m(s)\| ds \\ &\leq 2 \int_0^t [\gamma_T^{(1)} \sqrt{Z_m(s)} + \gamma_T^{(2)} (\sqrt{Z_{m-1}(s)})^N] \sqrt{Z_m(s)} ds \end{aligned}$$

$$\begin{aligned}
&\leq \gamma_T^{(2)} \int_0^T Z_{m-1}^N(s) \, ds + (2\gamma_T^{(1)} + \gamma_T^{(2)}) \int_0^t Z_m(s) \, ds \\
&\leq T\gamma_T^{(2)} \|w_{m-1}\|_{W_1(T)}^{2N} + (2\gamma_T^{(1)} + \gamma_T^{(2)}) \int_0^t Z_m(s) \, ds.
\end{aligned}$$

Using Gronwall's lemma, (2.35) leads to

$$(2.36) \quad \|w_m\|_{W_1(T)} \leq \mu_T \|w_{m-1}\|_{W_1(T)}^N,$$

where $\mu_T = 2\sqrt{\gamma_T^{(2)} T \exp((2\gamma_T^{(1)} + \gamma_T^{(2)})T)}$.

Choosing T small enough such that

$$\|u_1 - u_0\|_{W_1(T)} \mu_T^{1/(N-1)} = \|u_1\|_{W_1(T)} \mu_T^{1/(N-1)} \leq M \mu_T^{1/(N-1)} \equiv \beta_T < 1,$$

it follows from (2.36) that for all m and p ,

$$\begin{aligned}
(2.37) \quad \|u_m - u_{m+p}\|_{W_1(T)} &\leq (1 - \|u_1 - u_0\|_{W_1(T)} \mu_T^{1/(N-1)})^{-1} (\mu_T)^{-1/(N-1)} \\
&\quad \times (\|u_1 - u_0\|_{W_1(T)} \mu_T^{1/(N-1)})^{N^m} \\
&\leq (1 - \beta_T)^{-1} (\mu_T)^{-1/(N-1)} (\beta_T)^{N^m}.
\end{aligned}$$

Hence, $\{u_m\}$ is a Cauchy sequence in $W_1(T)$. Then there exists $u \in W_1(T)$ such that

$$(2.38) \quad u_m \rightarrow u \quad \text{strongly in } W_1(T).$$

Note that since $u_m \in W_1(M, T)$, there exists a subsequence $\{u_{m_j}\}$ of $\{u_m\}$ such that

$$(2.39) \quad \begin{cases} u_{m_j} \rightarrow u & \text{in } L^\infty(0, T; H_0^1 \cap H^2) \text{ weakly}^*, \\ \dot{u}_{m_j} \rightarrow \dot{u} & \text{in } L^\infty(0, T; H_0^1 \cap H^2) \text{ weakly}^*, \\ \ddot{u}_{m_j} \rightarrow \ddot{u} & \text{in } L^\infty(0, T; H_0^1) \text{ weakly}^*, \\ u \in W_1(M, T). \end{cases}$$

We have

$$\begin{aligned}
(2.40) \quad \|F_m(\cdot, t) - f(\cdot, t, u(t))\| &= \left\| \sum_{i=1}^{N-1} \frac{1}{i!} \frac{\partial^i f}{\partial u^i}(x, t, u_{m-1})(u_m - u_{m-1})^i \right\| \\
&\leq K_M(f) \sum_{i=1}^{N-1} \frac{1}{i!} \|u_m - u_{m-1}\|_{W_1(T)}^i.
\end{aligned}$$

Hence, (2.38) and (2.40) imply that

$$F_m(t) \rightarrow f(\cdot, t, u(t)) \quad \text{strongly in } L^\infty(0, T; L^2).$$

Finally, passing to limit in (2.2) and (2.3) as $m = m_j \rightarrow \infty$, there exists $u \in W(M, T)$ satisfying the equation

$$\langle \ddot{u}(t), w \rangle + \langle u_x(t) + \ddot{u}_x(t), w_x \rangle = \langle f(\cdot, t, u(t)), w \rangle,$$

for all $w \in H_0^1$ and the initial condition

$$u(0) = \tilde{u}_0, \quad \dot{u}(0) = \tilde{u}_1.$$

Uniqueness. Applying a similar argument as used in the proof of Theorem 2.1, $u \in W_1(M, T)$ is the local unique weak solution of problem (1.1)–(1.3).

Passing to the limit in (2.37) as $p \rightarrow \infty$ for fixed m , we get (2.29). In the same way as (2.29), (2.28) follows. Theorem 2.2 is proved completely. $\square$

Remark. (i) If the convergence of $\{u_m\}$ is only at a rate of order 1, it follows from (2.29) that the error at the m -th step is $C_T(\beta_T)^m$ with $0 < \beta_T = \mu_T < 1$ (T is small enough). If the convergence of $\{u_m\}$ is at a rate of order $N \geq 2$, this error is $C_T(\beta_T)^{N^m}$ and thus converges more rapidly, where $0 < \beta_T = M\mu_T^{1/(N-1)} < 1$ and T is also small enough.

(ii) In constructing a N -order iterative scheme, the function f has to satisfy (A2). This condition can be relaxed if we only consider the existence of a solution, see [10]–[13], [14], [17], [18].

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